TABLE OF CONTENTS

1.0 SCOPE

1.1 This Practice covers shop and field fabrication and inspection of piping assemblies, expansion joints, and furnace tube coil assemblies. Field erection and testing of piping systems is covered by EP 5-5-3, supplemental requirements for ASME I fabricated piping are covered in EP 5-5-8, and additional requirements for jacketed piping systems are covered in EP 5-5-5.

1.2 Selection of piping components and materials, other than furnace tube material, is covered in EP 5-2-1. Selection of furnace tube materials is covered in EP 8-2-1.

1.3 Supplemental requirements, including data to be furnished by the Purchaser, the Manufacturer's responsibility and documentation, and marking, packaging, and shipping for fabricated piping, are covered in EP 5-5-6.

1.4 Any deviation from this Practice shall be approved by the procedure described in EP 1-1-3.

1.5 An asterisk (*) indicates that a decision by the Owner's Engineer or Owner is required, or that additional information is furnished by the Purchaser.

1.6 A revision bar indicates the most recent changes made to this Practice.

1.7 There are supplemental requirements for piping in HF Alky Service. Refer to EP 5-10-2 for HF Alky Service.

2.0 REFERENCES

The following standards and publications are referred to herein.

Standards and Publications

Kern Engineering Practices	
EP 1-1-3	Deviations to the Kern Engineering Practices
EP 5-1-6	Piping Design to Avoid Vibration Fatigue Failure
EP 5-2-1	Selection of Piping Components and Materials
EP 5-2-2	Flanges, Gaskets, and Bolting
EP 5-2-3	Kern Engineering Piping Standards
EP 5-3-1	Valve Design and Selection Criteria
EP 5-4-1C	Metal Expansion Joints Inspection Checklist
EP 5-4-3	Supplemental Requirements for Expansion Joints
EP 5-5-1C	Piping Fabrication Inspection Checklist
EP 5-5-2	Welding Requirements for Piping Constructed to ASME B31.3
EP 5-5-3	Pipe Erection and Testing
EP 5-5-5	Supplemental Fabrication Requirements for Jacketed Piping
EP 5-5-6	Supplemental Requirements for Piping Fabrication
EP 5-5-7	Supplemental Requirements for Pneumatic Testing of New Piping Systems
EP 5-5-8	Supplemental Requirements for ASME I Piping
EP 5-6-2	Design and Installation of Piping Systems for Rotating Equipment
EP 5-10-2	Supplemental Requirements for Piping in HF Service
EP 8-2-1	Fired Heaters (API 560)
EP 10-2-1	Materials Requirements for Aggressive Environmental Services
EP 10-2-3	Material Hardness Requirements
EP 10-3-1	Shop and Field Coating
EP 11-1-1	Internal Insulating and Refractory Linings

Standards and Publications

EP 11-3-1	Insulation Design
EP 15-1-4	Positive Materials Identification (PMI)
ASME	
ASME I	Rules for Construction of Power Boilers
ASME B1.20.1	Pipe Threads, General Purpose (Inch)
ASME B16.5	Pipe Flanges and Flanged Fittings NPS 1/2 through NPS 24 Metric/Inch Standard
ASME B16.25	Buttwelding Ends
ASME B16.48	Line Blanks
ASME B31.1	Power Piping
ASME B31.3	Process Piping
ASME B31.4	Pipeline Transportation Systems for Liquid and Slurries
PFI	
PFI ES-3	Fabricating Tolerances
PFI ES-5	Cleaning of Fabricated Piping
PFI ES-7	Minimum Length and Spacing for Welded Branch Connections
PFI ES-24	Pipe Bending Methods, Tolerances, Process and Material Requirements

3.0 DEFINITIONS

3.1 Aggressive Environmental Service (AES) – As defined in EP 10-2-1.

3.2 Applicable ASME Piping Code – The ASME Piping Code which covers the piping systems being designed or fabricated. The ASME Piping Code will either be ASME B31.1, ASME B31.3, or ASME B31.4. The scope and coverage of each of these codes is stipulated in the Scope section of the code in question.

3.3 Company – Kern Energy.

3.4 Contractor – Party entering the contract with Kern Energy (to clearly define responsibility when multiple contractors are involved in a task, the responsible party will be explicitly called out in this specification, e.g., paint/coating supplier, steel supplier, fabrication contractor, transportation and installation contractor, or hook-up and commissioning contractor).

3.5 High Temperature Hydrogen Service – As defined in EP 10-2-1.

3.6 Hydrogen Service – As defined in EP 10-2-1.

3.7 Inspection Authority – An individual appointed or approved by the Owner who is responsible for the Owner's quality assurance program and has the knowledge and training to resolve issues relating to inspection and quality.

3.8 Inspector – The person or persons appointed by management to be responsible to the Inspection Authority for the inspection of equipment within the facility.

3.9 Manufacturer – The recipient of a direct or indirect Purchase Order for materials and/or equipment. In this context, a direct order is one issued to a Manufacturer by the Owner or Owner's representative. An indirect order is one issued to a manufacturer by the Manufacturer for materials, fabricated components, or subassemblies.

3.10 Owner – Kern Energy.

3.11 Owner's Engineer – A Kern Energy appointed engineer.

3.12 Owner's Representative – A Kern Energy appointed party able to issue a Purchase Order on behalf of Kern Energy.

3.13 Plant Inspection Authority – That person or body appointed by Kern Energy's Management to be responsible for examination and certification of equipment utilized within the petroleum refinery. Kern Energy Management may appoint more than one Plant Inspection Authority in order to cover specialized equipment such as rotating, instrumentation, and electrical equipment.

3.14 Purchase Order – The contractual document given to the Manufacturer to authorize a purchase.

3.15 Purchaser – The party placing a direct Purchase Order. The Purchaser is the Owner or Owner's Representative.

3.16 Vibration Service – Equipment and Service where the potential for vibration exists are defined in Table 1 of EP 5-1-6. Other equipment and services may be designated as Vibration Service by the Owner's Engineer. In addition to Table 1 of EP 5-1-6, the Owner's Engineer should refer to the Energy Institute, "Guidelines for the Avoidance of Vibration Induced Fatigue Failure in Process Pipework," latest edition, to determine other piping systems which may result as Vibration Service. A system is deemed not in Vibration Service (excluded from Table 1 of EP 5-1-6) when the quantitative Energy Institute guidance results in small bore piping LOF <0.3 and main line piping LOF <0.5.

4.0 DOCUMENTATION

4.1 (*) The inspection checklist for fabricated piping (EP 5-5-1C) or metal expansion joints (EP 5-4-1C) shall be completed by the Inspector prior to the start of fabrication.

4.2 Supplemental requirements and documentation for shop and field fabricated piping and expansion joints are covered in EP 5-5-6 and EP 5-4-3, respectively.

5.0 QUALITY CONTROL PROGRAM

The Manufacturer shall have as part of his usual business practice a quality control system. In addition, a quality control plan shall be developed for the manufacture of new fabricated piping to ensure that all technical requirements are followed. Requirements for the quality control system and plan for fabricated piping and expansion joints are covered in EP 5-5-6 and EP 5-4-3, respectively.

6.0 CODES, MATERIALS, AND GENERAL FABRICATION

6.1 (*) Pipe and piping components shall have all materials, design, and fabrication, including any examination and testing, meet all the requirements and limitations of ASME B31.3 or ASME B31.1 as applicable, except as modified in this Practice. In case of conflict among these requirements, the most stringent requirements, as determined by the Owners Engineer, shall govern.

6.2 Fabrication of boiler plant piping (steam, feedwater, and blowoff) within the jurisdiction of the ASME I shall comply with that code, all local regulations, all non-conflicting requirements of this Practice, and the supplemental requirements of EP 5-5-8.

6.3 (*) Materials used in the fabrication of piping shall be in accordance with the Company Piping Standards (refer to EP 5-2-3) that have been approved by the Owner's Engineer and which have been designated on drawings or line schedules for the specific piping that is to be fabricated.

6.4 (*) The conditions of design and the general arrangement of the piping, including the types of flanges and the dimensions of all parts, shall be specified in the Purchase Order.

6.5 (*) Unless otherwise specified herein, fittings or flanges differing from those specified in the Purchase Order shall not be substituted without the prior written approval of the Owner's Engineer.

6.6 (*) Piping shall be fabricated to form assemblies of the largest practicable sizes economically suitable for transportation to the job site. Written approval of these configurations shall be obtained from the Owner's Engineer prior to shipment.

6.7 (*) When specified in the Purchase Order, buttwelding end valves shall be furnished by the Purchaser for inclusion in fabricated assemblies.

6.8 (*) When a branch line to a header connection is of field fabricated size, the shop fabrication of the connection shall cease at the header fitting. Branch welding fittings, couplings, or reducing tees, as specified in the Purchase Order, shall be provided at all such connections.

6.9 Magnetic devices shall not be used to handle pipe or piping components.

6.10 (*) Field Fit Welds (FFW) shall be provided to ensure pipe spools can be aligned with equipment and other piping in the field without cutting and rewelding shop fabricated pipe spools. The installation procedure and potential errors in field location of equipment shall be considered when locating field welds. This is of special concern in piping to rotating equipment.

6.11 (*) Where Field Welds (FW) are required to join the ends of two shop fabricated spools to meet shipping dimensions or field erection of shop spools, the adjacent ends shall be beveled for field welding. Shop fabrication shall comply with the drawing dimensions. Unless otherwise specified by the Owner's Engineer, where Field Fit Welds (FFW) are required, one spool piece (at the FFW weld location) shall have a plain end six inches (150 mm) longer than the dimension indicated on the piping drawing. The adjacent spool piece shall have a beveled end and shall be fabricated to the indicated dimension.

6.12 Fabrication requirements and details for jacketed piping systems are covered in EP 5-5-5.

7.0 DIMENSIONAL TOLERANCES

7.1 Pipe fabrication tolerances shall be in accordance with PFI ES-3, except that:

7.1.1 The flanges mating to rotating equipment shall be per EP 5-6-2. These limits also apply to all flanged joints made with non-ductile materials (for example, cast iron and glass lined pipe).

7.1.2 ASME B31.3 flange face alignment tolerances apply for applications other than those listed above.

7.1.3 The lateral offset of flanges in any direction from the specified direction shall not exceed 1/16 inch (1.5 mm).

7.1.4 For Category D, Normal Fluid Service, Category M, Severe Cyclic, Temperature > 850°F (455°C), or Rating ≥ Class 900, flange faces shall be aligned with 0.03 inches (0.8 mm) measured across the outside of the diameter of the flange face at any location around the flange.

7.1.5 The tolerances on linear dimensions of spool pieces composed of piping components with intervening pipe shall not exceed: 1/8 inch for NPS 10 and smaller (3 mm for DN 250 and smaller), 3/16 inch for NPS 12 through NPS 24 (5 mm for DN 300 through DN 600), and 1/4 inch for greater than NPS 24 (6 mm for greater than DN 600). Tolerances shall not be cumulative.

7.1.6 (*) For spool pieces composed entirely of piping components with no intervening pipe, the sum of the tolerances on linear dimensions for the components could exceed the requirement of above. In such cases, the following procedure shall be used to meet the tolerances on linear dimensions:

1) The fabricator shall first make every reasonable effort to meet the requirement of the previous paragraph using piping components from their stock.
2) (*) If piping components cannot be found in-stock such that the requirement of the previous paragraph is met, the fabricator shall adjust the linear dimensions of adjacent spool pieces such that the assembly will meet the requirements of the previous paragraph. In cases where adjustments are made to the linear dimensions of adjacent spool pieces in accordance with this paragraph, spool drawings shall list these adjustments in the notes.
3) (*) In cases where the procedures in the two previous paragraphs cannot be used to meet the requirement of the previous paragraph, and a tight tolerance shall be maintained, piping components may be machined. Such cases require review and approval by the Owner's Engineer. However, in no case shall flange faces be machined in order to meet tolerance on linear dimensions.

7.2 Heating of pipe to satisfy the dimensional tolerances specified above shall conform to the following requirements:

1) Only carbon and low alloy steel (P-1, P-4, and P-5) materials may be heated to facilitate fabrication.
2) Pipe that has not been postweld heat treated may be heated for minor corrections in fit. Under no circumstances, however, may pipe materials be quenched to correct alignment. For carbon steel that does not require PWHT, the maximum temperature during alignment corrections shall be 1200°F (650°C), with the following:
   - Temperature-indicating crayons or contact thermometers shall be used to measure the maximum temperature.
   - Forcing may be applied, if necessary.
   - Cooling in still air shall be applied.
3) (*) Unless otherwise specified by the Owner's Engineer, the maximum temperature used to correct alignment is 1200°F (650°C).
4) (*) Pipe that has been postweld heat treated shall not be heated to correct misalignment without approval of the Owner's Engineer. If approval is given, heating shall be done with electric resistance heaters. After any heating to correct misalignment, hardness shall not exceed 225 HB in any part of the heated area. Heating shall be done over a 24 inch (600 mm) length, minimum. Heaters shall be fully insulated. Insulation shall extend a minimum distance of 24 inches (600 mm) on either side of the heaters. Insulation shall not be removed until all of the pipe has cooled to below 600°F (315°C).

7.3 The ends of piping components to be joined shall be aligned as accurately as is practicable within existing commercial tolerances on diameter, wall thickness, and out-of-roundness. Where the internal misalignment between piping components exceeds 1/16 inch (1.5 mm), the component with the wall extending internally shall be trimmed in accordance with the applicable ASME Piping Code. This misalignment tolerance is not permitted in piping to and from reciprocating compressors (through the second change in direction) or in piping with Severe Cyclic Conditions, as defined in ASME B31.3. For such services, the inside diameter of the pipe should be machined or ground to a close tolerance and the root pass should be welded by the tungsten inert gas process.

7.4 The maximum allowable gap between reinforcing pads and the curvature of the pipe shall not exceed 1/16 inch (1.5 mm).

8.0 ADDITIONAL TOLERANCE REQUIREMENTS FOR LARGE DIAMETER PIPE

8.1 (*) The requirements of this section shall be used for all shop and field fabricated pipe joints of NPS 36 (DN 900) and larger, unless otherwise specified by the Owner's Engineer.

8.2 Field joint design for large diameter fabricated pipe ends shall be one of the types specified in Table 1.

8.3 Truing rings shall be considered temporary if installed on pipe with a design metal temperature greater than 400°F (205°C). Temporary truing rings shall be stitch welded to the pipe with 2 inch (50 mm) welds on 8 inch (200 mm) centers, marked in accordance with Section 21.0, and removed prior to hydrotest.

8.4 Truing rings can be considered permanent if installed on pipe with a design metal temperature of 400°F (205°C) or less. Permanent truing rings shall be continuously welded to the pipe.

8.5 Truing rings shall be a minimum of 1/2 inch (13 mm) thick by 3 inches (75 mm) in width. Temporary truing rings shall be fabricated from carbon steel. Permanent truing rings shall be fabricated from the same material as the pipe to which they are attached.

8.6 Truing ring tolerances shall be as follows:

1) Outside diameter: Based on circumferential measurement ± 0.25% of the specified outside diameter.
2) The out-of-roundness: Difference between major and minor outside diameters shall be 0.5%.

9.0 BENDS, ELBOWS, AND MITER BENDS

9.1 (*) Use of pipe bends and miter joints shall be approved by Owner's Engineer, except pipe bends may be used for small diameter piping NPS 1 (DN 25) and smaller in place of elbows. Centerline radius of smooth pipe bends shall be equal to at least five times the nominal pipe diameter.

9.2 If the pipe bend contains a longitudinal weld, this weld shall be located in the neutral zone of the bend.

9.2.1 If the bend will be installed in the horizontal plane, the longitudinal weld needs to be located on the top of the pipe.

9.2.2 If one spool is bent in various planes, the longitudinal weld needs to be 45 degrees from the top of the pipe.

9.2.3 No girth weld shall be located in the bend.

9.3 The specified mechanical properties of the original material specification (mechanical, metallurgical, corrosion resistance) shall be met after cold bending or hot bending and after any heat treatment.

9.4 If high frequency induction bending is used after heat treatment, the specified mechanical properties of the original material specification shall be met.

9.5 Miter bends shall comply with the following:

9.5.1 Bends exceeding 45 degrees shall be at least three pieces (two miter cuts), with not less than eight times the pipe wall thickness between the centerlines of the welds at the crotch.

9.5.2 Bends shall have a centerline radius at least equal to the nominal pipe size.

9.5.3 The maximum miter angle (half the change of direction) shall be 22-1/2 degrees up to 200 psig (1.5 MPaG) design pressure, 15 degrees up to 400 psig (3 MPaG), and 11-1/4 degrees over 400 psig (3 MPaG).

9.6 (*) Butt welding elbows shall be of the long radius type, unless otherwise approved by Owner's Engineer.

9.7 The following requirements shall be followed for bending and forming of pipe bends:

9.7.1 Pipe bends shall be per PFI ES-24 and the requirements of this Practice.

9.7.2 The ovality of a pipe bend shall not exceed the minimum of that listed below:

1) 2% maximum (internal or external pressure) for severe cyclic service or temperature > 850°F (455°C).
2) 8% maximum for internal pressure and Category D, Normal Fluid Service, Rating ≥ Class 900, or Category M service.
3) 3% maximum for external pressure and Category D or Normal Fluid Service.
4) Maximum allowed by governing code.

9.7.3 (*) Bending and forming procedures, including preheat and postheat treatment, shall be approved by the Owner's Engineer prior to use. These procedures shall include a description of all heat treatment, control of temperatures, and material hardness verification. Uniform heating of the materials is required before forming and hot bending.

9.7.4 Bends shall only be made from the seamless and electric fusion welded pipe listed in EP 5-2-1.

9.7.5 Rotary draw cold bending is performed on a bending machine, which utilizes a forming die, ball or plug mandrel (when required), pressure die, and wiper die. The bending of material takes place at ambient temperature. The following requirements for rotary draw cold bending shall be followed:

9.7.5.1 The wall thickness shall be tested prior to shipment by the pipe supplier, to ensure the estimated thickness after bending will not be less than the minimum required wall thickness calculated for pressure and temperature.

9.7.5.2 (*) Upon completion of the ending and trimming, the substantial removal of the lubricant shall be achieved through the use of steam or hot water. Other methods shall be approved by Owner's Engineer.

9.7.5.3 The following tolerances on linear dimensions shall be met. Dimensions taken from the center of bends to the ends:

1) Specified tangent lengths:
   - ± 1/8 inch (3 mm) for specified tangent lengths up to 5 feet (1.5 m).
   - ± 1/4 inch (6 mm) for specified tangent lengths more than 5 feet (1.5 m).
2) Centerline radius:
   - ± 1/2 inch (13 mm) for centerline radius up to 24 inches (600 mm).
   - ± 1 inch (25 mm) for centerline radius more than 24 inches (600 mm).
3) Degree of bend tolerance:
   - ± 1 degree.
4) Center to center tolerance for 180 degree bends:
   - ± 1/4 inch (6 mm).
5) Roundness of end:
   - ± 1% when tangent length is at least 1-1/2 D.
   - ± 2% when tangent length is less than 1-1/2 D.

6) Center-to-center dimension on offset bends:
   - ± 1/4 inch (6 mm) for 1 inch (25 mm) OD to 6-5/8 inch (168 mm) OD pipe or tubing.
   - ± 1/2 inch (13 mm) for pipe or tubing greater than 6-5/8 inch (168 mm) OD.

9.7.6 Pipe bends with the following defects will be rejected:

1) The pipe wall thickness at the thinnest point after bending, less corrosion allowance, shall not be less than the calculated thickness required for pressure and temperature.
2) Wrinkles and bulges after bending and heat treatment.
3) Cracks and/or laminations.
4) Dimples or craters ≥ corrosion allowance.

9.7.7 Bends of ferritic materials shall have a radius measured to the centerline of the pipe of at least five times the nominal pipe diameter. Bends in NPS 14 (DN 350) and larger standard wall and lighter pipe shall have a radius of at least six times the nominal pipe diameter.

9.7.8 Carbon and 1/2% to 9% chromium molybdenum steels shall not be hot bent at temperatures exceeding 1900°F (1040°C). After hot bending, if the hardness of carbon steel or 1/2 percent chromium-molybdenum steels exceeds 225 Brinell hardness number (HB), the bends shall be stress relieved. Hot bends in 1% to 9% chromium-molybdenum steels shall be normalized and drawn or annealed. The hardness of all bends after heat treatment shall not exceed 225 HB. (NOTE: The mechanical properties of controlled rolled, microalloyed, and cold expanded pipe can be significantly changed by hot bending and/or normalizing).

9.7.9 (*) Cold-formed bends in chromium-molybdenum materials shall be stress relieved. Cold-formed bends in carbon steel materials shall be stress relieved, unless approved by the Owner's Engineer.

9.7.10 Cold bends of carbon steel shall not be used below -20°F (-29°C), unless they are normalized at a temperature of 1650°F ± 50°F (900°C ± 28°C) for one hour.

9.7.11 Bending or forming is not permitted after heat treatment, unless followed by reheat treatment.

9.7.12 (*) The induction bending process shall only be used when approved by the Owner's Engineer.

9.7.13 Water quenching of non-austenitic steels is prohibited, except in the induction heat bending process. Acceptance requires that one test bend per size and weight be produced immediately prior to the production bends. The test bends shall be examined to verify that thicknesses, ovality, and mechanical properties are within specifications. If the induction heat bending process is used for production bends, hardness measurements shall be made every 12 inches (300 mm) on both the inside radius and outside radius of each bend. If any hardness measurement on a bend exceeds 225 HB, stress relief of that bend is required. To ensure acceptable hardness limits for low alloy steel, stress relief per EP 5-5-2 shall be required of all P3, P4, and P5 materials.

9.7.14 Prior to hot bending and shipment, all austenitic stainless steel materials shall be free from all markings, such as stamping, grease, lead, soap, or sulfur-containing compounds. Stainless steel metal tags or noncorrosive paints shall be used for identification purposes.

9.7.15 (*) Cold and/or hot bends of materials other than P1, P2, P4, or P5 material shall be reviewed and approved by the Owner's Engineer. Involve the Owner's Engineer after a couple of attempts have been made to resolve any non-conformance issues stipulated above.

10.0 SOCKET WELD FITTINGS

10.1 Socket welding is preferred for diameters NPS 2 and less, as specified within the piping line class.

10.2 Socket weld fitting assembly shall have the pipe or tube inserted into the socket to the maximum depth then withdrawn no less than 1/16 inch (1.6 mm) and no more than 1/8 inch (3.2 mm) away from the contact, between the end of the pipe and the shoulder of the socket. See Figure 1D.

11.0 BRANCH CONNECTIONS

11.1 (*) The preferred branch connections are shown in Table 2A and listed in the piping classes. With Owner's Engineer approval, any additional alternative branch connections shown in Table 2B may be used. For vibration service, additional requirements for branch connections are provided in EP 5-1-6.

11.2 Integrally reinforced branch welding fittings larger than NPS 2 (DN 50) (Weldolet, Latrolet, Elbolet), which abut the outside surface of stainless steel or nickel based alloy pipe with a schedule 5S or 10S thickness, shall be specified to be a schedule 10 design. The use of a standard weight or schedule 40 design is not permitted.

11.3 (*) Fabricated branch connections shall be directly joined to the header with welds, providing full penetration of the branch to header. Unless otherwise specified by the Owner's Engineer, the branch connection joint configuration shall preferably use a "set-in" rather than "set-on" type design; see Figure 2. All branch connections shall be visually inspected during fabrication, both internally, when possible, and externally, in addition to all other types of examination that may be specified. Reinforcing pads, when required, shall be added as a subsequent fabrication step after visual inspection.

11.4 Reinforcement for pipe-to-pipe branch connections shall be designed in accordance with the applicable ASME Piping Code. All reinforcement pads for pressure openings, or each segment of built-up type reinforcement pads for pressure openings, shall be provided with one NPT 1/4 inch (AN 6) hole for testing and venting. Reinforcing pads for structural attachments and trunnions shall be provided with one NPT 1/4 inch (AN 6) hole in each segment for venting. The vent shall not be at apex of curvature. Each reinforcing pad or each segment thereof shall be tested at 15 psig (100 KPaG), with dry air and a soap solution before hydrotest. Remove test fitting and pack the test port with grease.

11.5 Fabricated branch connections, other than welding tees or those shown in Figure 3, shall not be used for severe cyclic conditions as defined in ASME B31.3 or in vibratory service (e.g., reciprocating compressor or pump).

11.6 Fabrication details for couplings shall be in accordance with Figure 4.

11.7 Fabrication details for branch welded-on fittings (weldolets) shall be in accordance with Figure 5A and Figure 5B.

11.8 (*) Full-sized or reduced branches at angles other than 90 degrees shall not be employed, except when required, because of flow and pressure drop considerations (for example, in flare lines and lines with flow induced vibration concerns). The angle between the branch and the run of the header shall not be less than 45 degrees. It is preferred that the centerline of the branch intersect the centerline of the header. Exception: Where flow conditions control and the Owner's Engineer has given prior approval, the branch may intersect the header in a tangential or semi-tangential direction, provided the angle between the branch and the run of the header is not less than 60 degrees and the intersecting surfaces form a welding angle of not less than 45 degrees. Forged laterals are preferred and can be designed per the M.W. Kellogg "pressure area" method. Alternatively, integrally reinforced latrolets may be used. With Owner's Engineer approval, fabricated laterals can be used, but they require 100% replacement of the reinforcement area (no credit for excess metal from the run or branch).

11.9 (*) Ribs and gussets shall not be used for pressure strengthening of branch connections without prior approval of the Owner's Engineer. Use of ribs, gussets, clamps, and plates to provide strength for external and thermal loads may be considered with prior approval of the Owner's Engineer.

11.10 Socket-welded branch connections shall not be used in service where crevice corrosion or erosion is likely to occur.

11.11 (*) Branch connections in "Pulsation Bottles" for reciprocating compressors shall be made by the use of welded-in contour inserts, extruded welding tees, or other methods that produce connections that have equal (or better) capability to withstand vibration and pulsation. All designs shall be approved by Owner's Engineer. Bottles with multiple nozzles which connect to compressor cylinders shall be shipped with only one of these nozzles welded in. The other nozzle(s) shall be shipped loose for field fit-up between the compressor and the bottle.

11.12 Where a line with a lower rating connects to pipe or equipment with a higher rating, the line shall have the higher pressure rating up to and including the first block valve, control valve, or check valve, up to and including the second valve and the bleed valve when double block valves are used, and up to and including the block valve and check valve when both a block valve and check valve are used. For alloy lines, the alloy material may terminate with the first block valve, which shall be alloy.

11.13 All small piping connections less than or equal to NPS 1-1/2 (DN 40) shall be fabricated using a minimum length of pipe between the takeoff point and valve, and without the use of elbows between the valve and takeoff point.

12.0 FLANGES, BLANKS, UNIONS, AND VALVES

12.1 All flanges shall be in accordance with EP 5-2-2.

12.2 (*) Unless otherwise approved by the Owner's Engineer, slip-on flanges in Hydrogen Service and High Temperature Hydrogen Service shall have the enclosed spaced between the outside diameter of the pipe and the bore of the flange vented to atmosphere by a 1/8 inch (3 mm) diameter hole. Such flanges shall be predrilled before assembly.

12.3 Flange bolt holes shall straddle the established centerlines (horizontal, vertical, or layout centerlines), unless the flange shall be rotated to match the bolt holes in flanges on equipment such as pumps.

12.4 (*) Slip-on flanges shall be positioned such that the distance from the face of the flange to the pipe end shall be a minimum of the nominal pipe wall thickness plus 1/8 inch (3 mm), and a maximum of the nominal pipe wall thickness plus 3/8 inch (10 mm). Slip-on flanges shall be welded inside and outside, with full fillet welds using a minimum of two weld passes; see ASME B31.3, Figure 328.5.2B(1). Unless otherwise specified by Owner's Engineer, slip-on flanges shall not be directly welded to pipe fittings (e.g., welding elbows and tees).

12.5 (*) Welding neck flanges shall be used where flanges are attached directly to welding fittings (even though slip-on flanges are specified in the piping material specification), unless otherwise approved by Owner's Engineer and shown on piping drawings.

12.6 Welding neck orifice flanges shall be the same bore as the pipe to which they are attached and shall be aligned as accurately as is practicable.

12.7 Flanges in refractory lined pipe shall be per EP 11-1-1.

12.8 Piping line blanks shall be per ASME B16.48, latest edition.

12.9 Pipe unions are not acceptable except per the provisions stipulated in EP 5-2-1.

12.10 Valve selection and design shall be per EP 5-3-1.

12.11 Welding is not permitted on valves equipped with soft seats, unless seats have been removed or precautions (recommended by the valve Manufacturer) have been taken to prevent damage.

12.12 Bracing requirements of small diameter valves in vibration service are covered in EP 5-1-6.

13.0 INSULATION AND REFRACTORY

13.1 (*) When specified by the Owner's Engineer, insulation supports shall be installed per EP 11-3-1.

13.2 Installation of refractory linings shall be per EP 11-1-1 when internal linings are specified in the Purchase Order.

14.0 PIPE JOINT DESIGN

14.1 Piping in all services shall be constructed as shown in Table 3.

14.2 All welding shall be in accordance with Section 15.0.

14.3 Taper pipe threads (NPT) per ASME B1.20.1 shall be used for all threaded pipe joints, including thermowell nozzles.

14.4 Socket weld fittings shall be per EP 5-2-1.

15.0 WELDING

15.1 All welding procedures and welding shall satisfy the requirements of the applicable ASME Piping Code, EP 5-5-2, and the additional requirements of this Practice.

15.2 All longitudinal, girth, miter, and branch connection welds shall have 100% penetration of the joint.

15.3 (*) Weld ends for beveled pipe shall be prepared per ASME B16.25, except that the bevel angle may not be less than 30 degrees without approval from the Owner's Engineer.

15.4 Radial misalignment at the joining ends of piping components shall be limited to 1/8 inch (3 mm) or 1/4 the pipe wall thickness, whichever is less. Internal radial misalignment exceeding 1/16 inch (1.5 mm) shall be taper trimmed so that the adjoining internal surfaces do not exceed 1/16 inch (1.5 mm); however, the resulting thickness of the welded joint shall not be less than the minimum design thickness, plus corrosion allowance. A 3 to 1 taper shall be used.

15.5 (*) The clear distance between centerlines of adjacent girth butt welds shall not be less than six times the pipe wall thickness, or 3 inches (75 mm), whichever is greater. With Owner's Engineer approval, the distance between centerlines of welds can be less, given the edge-to-edge distance of the welds is at least four times the pipe wall thickness, to avoid overlapping of weld residual stresses.

15.6 (*) The clear distance between attachment welds for adjacent branch connection fittings shall be based on the criteria stipulated in PFI ES-7. For NPS 2 (DN 50) and smaller branch connections, which are not covered by PFI ES-7, the minimum spacing from the edge of branch connections (dimension "C" in PFI ES-7) shall be 3 inches (75 mm). With Owner's Engineer approval, these spacing requirements may be reduced. For such reduction, consideration shall be given to a change in the branch-to-header flexibility and local stresses, along with required changes to shop operations to prevent distortion of the header.

15.7 Longitudinal seams in adjoining lengths of welded pipe shall be staggered and shall be so located as to clear openings and external attachments. Clearance and stagger shall not be less than ten times the nominal pipe wall thickness or 90 degrees, whichever is greater. Longitudinal seam welds shall be located in the top quadrant of the pipe when practicable. Shop fabricated seamed piping shall be inspected for long seam alignment on adjoining spools to meet staggered long seam expectations of this specification.

15.8 (*) Dissimilar Metal Welds (DMW): Consult the Owner's Engineer for applications where a field dissimilar weld is being considered.

15.8.1 DMWs should be performed in a shop. Fabrication should be conducted such that field welds are similar metal welds only, whenever practical. For example, when a field transition butt weld between carbon steel pipe and alloy or non-ferrous alloy pipe is required, a 12 inch (300 mm) long section of carbon steel pipe of the required diameter and wall thickness shall be shop welded to the alloy steel or non-ferrous pipe so that a similar material weld can be made in the field.

15.8.2 For high temperature applications, nickel-based filler metals should be used for DMWs. 300 series fillers should not be used above 600°F (315°C).

15.8.3 Where shop DMW welds are not practical, buttering of the ferritic side of the joint should be considered, followed by PWHT, prior to the completion of the DMW. On buttered joints, the thickness of the weld metal should be no less than 0.25 inches (6.4 mm) after machining.

15.8.4 In steam generating equipment, the weld at the high temperature end should be made in the penthouse of header enclosure, outside of the heat transfer zone.

15.8.5 For high temperature installations, consider installing a pup piece that has an intermediate thermal expansion coefficient between the two materials to be joined.

15.9 When field welds are added in lines which pass through building walls and roofs, column lines of structures, foundations, platforms, or grade elevation, the welds shall be located at least 12 inches (300 mm) outside of or above such structural items, when practical.

15.10 Tack welding butt joints for completion at a later date shall be in accordance with EP 5-5-2.

15.11 All piping sections to be lined with castable refractory shall either be match marked and provided with truing rings per Section 8.0, prior to the installation of lining, or the lining shall terminate 8 inches (200 mm) from the pipe end to facilitate field fitup and welding. Hammering or deforming fully lined pipe (i.e., lined to the end of the pipe) to affect joint fitup with other pipe is not acceptable.

15.12 Sizes of fillet welds for attaching socket-welding components and slip-on flanges shall be not less than shown in the applicable ASME Piping Code, and shall also comply with the requirements below:

1) The weld shall be at least a two-pass weld.
2) The external fillet weld attaching the hub of the slip-on flange to the pipe shall have a weld leg dimension no less than 1/4 inch (6 mm).
3) The internal fillet weld attaching the slip-on flange to the pipe shall have a weld leg dimension equal to the nominal thickness of the pipe.

15.13 Internal welds of piping which is to be internally coated (with "Scotch-Kote" or similar material) or piping in turbulent sensitive services (e.g., acid service) shall be ground smooth and flush with the ID of the pipe. If this cannot be done, GTAW root pass welding using an argon gas back purge shall be used, and the root pass shall be as flat or smooth as possible.

15.14 Excessive weld projections, as determined using the applicable ASME Piping Code, on accessible joints shall be removed by grinding. Welds having excessive projections on inaccessible joints shall be cut and re-welded.

15.15 Field welded piping "tie-ins" for low alloy steels (P-4, P-5A, P-5B, and P-5C), stainless steels (P-7 and P-8), and all compressor piping shall have the root pass welded with GTAW, with the backside of the weld purged with argon gas.

15.16 Seal welding requirements are covered in EP 5-5-3.

16.0 HEAT TREATMENT

16.1 Postweld heat treatment (PWHT) shall be in accordance with the requirements of EP 5-5-2 and EP 10-2-1.

16.2 (*) Approval of preheat and postweld heat treatment procedures is required by the Owner's Engineer for the following:

1) Material with thicknesses greater than 2 inches (50 mm).
2) Any annealing procedures.
3) Welds joining dissimilar materials.
4) Materials not covered by the applicable ASME Piping Code.

16.3 It shall be the Manufacturer's responsibility to demonstrate that the postweld heat treatments selected to meet the Brinell hardness requirements of EP 10-2-1 and EP 10-2-3 will not reduce material strength levels below code allowances.

16.4 (*) Exposed machined and threaded surfaces shall be protected from oxidation during heat treatment. Materials and procedures to be used to provide this protection shall be approved by the Owner's Engineer.

16.5 When fabricated piping requires heat treatment, all welded attachments, such as pipe supports, insulation supports and gussets, and reinforcing pads shall be installed prior to final heat treatment. In addition, all temporary welded attachments shall be removed before final heat treatment. Subsequent welding or heating after final heat treatment is strictly prohibited.

17.0 CLEANING OF PIPING

17.1 (*) Upon completion of fabrication and prior to testing or shipment, the Manufacturer shall clean the outside and inside of all piping by suitable means, such as wire brushing, to ensure the surfaces are free of all loose scale, dirt, welding slag and flux, weld rod stubs, weld cuttings, ships, and other debris. The assembly shall then be blown out with compressed air and the inside inspected for cleanliness.

17.2 Wire brushes used on austenitic stainless steel shall be austenitic stainless steel and shall not have been used on any other materials. Any necessary additional cleaning and the extent of piping to be cleaned shall be as specified by the Purchaser.

17.3 Wire brushes used on titanium shall be austenitic stainless steel, shall not have been used on any other materials, and shall be thoroughly cleaned before use.

17.4 For shop fabricated piping that is to be hydrotested, cleaning of all components after testing and prior to shipping is required.

17.5 Cleaning of fabricated piping shall be per PFI ES-5.

17.6 (*) Cleaning of titanium piping requires special consideration. Consult the Owner's Engineer for direction.

18.0 INSPECTION AND TESTING

18.1 General

18.1.1 All material and fabrication are subject to inspection by the Inspector. Rejections by the Inspector are final. The inspection by the Inspector does not relieve the fabricator (erector, Manufacturer, or Vendor, as appropriate) of his responsibility for complying with the Purchase Order.

18.1.2 The fabricator shall advise the Inspector at least five working days before starting and completing fabrication, hold points, specified testing, and preparation for shipment.

18.1.3 The examination of pipe welds shall be in accordance with the applicable ASME Piping Code and the additional requirements of this Practice. For jacketed piping, the inner pipe welds shall be examined prior to assembly of the outer pipe.

18.1.4 Chisel marks, clamp marks, and notches shall be blended to smooth contours, with pointed or sharp corners or bottoms removed. Dents, notches, or marks which are deeper than 1/8 inch (3 mm), greater than 12-1/2% of the nominal wall thickness, or which reduce wall thickness below the minimum required design thickness plus corrosion allowance shall not be accepted and shall be repaired.

18.1.5 (*) The procedure used for the visual examination shall be compliant with the requirements of ASME V – 2021 Article 9. This procedure shall be reviewed and approved by the Owner's Engineer prior to the start of visual examination.

18.1.6 Socket-welded fitting acceptance criteria include verification of appropriate fitting preparation and of the minimum/maximum end gap, both performed prior to welding, and verification of proper fillet weld leg/throat size and minimum 3 weld passes (2 visible) for fittings that are 1 inch and larger, both performed after welding.

18.1.7 In-process visual examination shall not be substituted for any required radiographic or ultrasonic examination. All welds shall be visually inspected.

18.1.8 Weld examination shall begin as soon as fabrication has started.

18.1.9 Welds that are deposited by procedures or welders differing from those properly qualified shall be completely removed and redone correctly.

18.1.10 Maximum allowable projections of weld metal into the pipe bore at welded butt joints shall be per the applicable ASME Piping Code and the following:

1) Plastic or elastomeric lined pipe. Internal porous areas, cavities, and pockets shall be filled with weld metal. All internal welds shall be ground to a smooth, uniform surface.
2) Orifice flanges and "meter tubes." Welds joining orifice flanges to pipe or "meter tubes" to pipe or fittings shall have an ID smooth and flush with the ID of the pipe.
3) Internal welds of the intake and interstage piping for reciprocating and rotary screw compressors shall be ground smooth and flush with the ID of the pipe. If this cannot be done, the root pass shall be made by an inert gas welding process and the underside of the weld protected from oxidation by inert gas purging during welding.

18.1.11 Delayed cracking inspection shall be performed for the following steels whenever heating for dehydrogenization is not implemented before cooling down, or when a weld will reach a temperature below 200°F (93°C) before PWHT:

1) CrMo steels (e.g., 1.25Cr-0.5Mo, 2.25Cr-1Mo, 5Cr-0.5Mo, 9Cr-1Mo).
2) Low alloy air hardenable steels.
3) Martensitic or ferritic stainless steels.
4) Carbon steels (CE ≥ 0.45 % and thicker than 0.75 inch (19 mm).
5) Carbon steels with UTS ≥ 70 ksi (483 MPa).

18.1.12 Delayed cracking inspection shall be carried out no sooner than 48 hours after completion of welding, unless GTAW is used as the welding method for the entire weld. Where GTAW is the used welding method, NDE shall be carried out no sooner than 16 hours following the completion of welding.

18.2 Radiographic Examination

18.2.1 Weld radiography shall be in accordance with the applicable ASME Piping Code and the additional requirements of this Practice.

18.2.2 Circumferential (girth) butt and miter groove welds shall be examined by full radiography (complete circumference of weld seam) to the inspection frequency and acceptance criteria in Table 4. The inspection frequency and acceptance criteria may be modified as follows:

1) (*) The inspection frequency and/or acceptance criteria may be increased, based on special service requirements by the Refinery Inspection Authority.
2) (*) The inspection frequency and/or acceptance criteria may be adjusted, contingent upon approval of the Refinery Inspection Authority. However, in no case shall the adjusted inspection requirements be less than those required by ASME Piping Code.

18.2.3 Longitudinal welds in pipe fabricated to API and ASTM specifications shall be examined per the applicable API/ASTM specification.

18.2.4 (*) Ultrasonic examination shall not be substituted for radiographic examination, unless approved by the Refinery Inspection Authority.

18.2.5 In addition to the 100% examination requirements in Table 4, the following welds shall be 100% radiographed (complete circumference of weld seam), using the acceptance criteria in Table 4. These welds shall not be included in the weld count used to establish the random radiography (less than 10%) examination requirements in Table 4.

1) (*) All tie-in butt welds to existing piping. When approved by the Refinery Inspection Authority, spot radiography may be substituted for full radiography for piping systems in Category D Fluid Service.
2) All welds made to branch contour inserts (sweep-o-lets).
3) All circumferential, longitudinal, and helical (spiral) groove welds in piping subject to a Pneumatic Leak Test per ASME B31.3, paragraph 345.5 or a Hydrostatic-Pneumatic Leak Test per ASME B31.3, paragraph 345.6; see EP 5-5-3 for leak testing requirements.
4) All circumferential, longitudinal, and helical (spiral) groove welds in piping subject to an Alternative Leak Test per ASME B31.3, paragraph 345.9; see EP 5-5-3 for leak testing requirements.
5) Category M Fluid Service.

18.2.6 When the minimum percentage of welds subject to radiographic examination is less than 100% per Table 4, the welds to be examined by random radiography shall be identified by the piping fabrication inspector. The piping fabrication inspector is defined in the ASME Piping Code (for example, see paragraph 340.4 in ASME B31.3). The method of selection shall be based on the following:

1) All personnel involved in the fabrication and welding of piping shall have no prior knowledge that a weld is to be radiographed until after completion of the weld.
2) Each welder's output shall be nominally examined to the frequency stipulated in Table 4. The welds selected for radiographic examination shall be evenly distributed, taking into account the different weld procedures and pipe sizes of welds made by each welder.
3) At least one weld for each weld procedure specification (WPS) made by each welder shall be radiographed.
4) Welds that are examined to the 100% radiography requirements of Table 4 shall not be included in the weld count used to establish the required random radiography percentages in this table.

18.2.7 In addition to the requirements in Table 4, for weld categories that require less than 100% radiographic examination, fabricated piping with a diameter greater than or equal to NPS 24 (DN 600) shall be examined by spot radiography, in accordance with the following criteria:

1) All circumferential (girth) weld seams in pipe NPS 24 (DN 600) and larger shall be examined in one location.
2) All circumferential weld seams in pipe larger than NPS 36 (DN 900) shall be examined in two locations.
3) (*) The location of the spot radiograph shall be randomly selected, but shall include all "T" intersections between circumferential and longitudinal welds or designated to a specific location by the Inspector.
4) The film length for each spot radiograph shall be 17 inches (425 mm).
5) The acceptance criteria for spot radiographs shall be per Table 4.
6) (*) Ultrasonic examination may be substituted for spot radiography when approved by the facility Inspection Authority. The required examination area shall encompass a minimum of 17 inches (425 mm) of the weld seam.

18.2.8 The sequence of radiographic examination for final acceptance of piping welds which require post weld heat treatment (PWHT) shall be in accordance with the following:

1) For piping systems constructed of carbon steel that have a wall thickness that is less than or equal to 0.75 inches (19 mm), radiographic examination may be performed before or after the final PWHT.
2) For piping systems constructed of carbon steel with a wall thickness that exceeds 0.75 inches (19 mm) and all other alloy materials which require PWHT, radiographic examination shall be performed after the final PWHT.

18.2.9 5% of the seal welds of threaded joints shall receive a surface examination of the final pass.

18.3 Ultrasonic Examination

18.3.1 Ultrasonic examination shall be in accordance with the applicable ASME Piping Code, the additional requirements of this Practice, and EP 5-5-2, paragraph 6.3.4.

18.3.2 (*) Ultrasonic testing shall be specified for all groove welds containing more than one pass or layer of short-circuiting GMAW. Radiography is not acceptable. Inspection frequency and acceptance limits shall be specified by the Owner's Engineer. The Vendor shall demonstrate his abilities to resolve cold lap with ultrasonic testing on suitable calibration standards.

18.3.3 (*) In addition to the required radiographic inspection, 20% of all welds where the pipe wall thickness exceeds 1-1/4 inches (32 mm) for carbon steel, and 3/4 inches (19 mm) for low alloy steel (P-3, P-4, and P-5), shall be 100% ultrasonically inspected.

18.3.4 Branch connections that are not pressure tested shall be inspected 100% by ultrasonic examination.

18.4 Magnetic Particle and Liquid Penetrant Examination

18.4.1 (*) Magnetic particle and liquid penetrant examination and acceptance criteria shall be performed in accordance with the applicable ASME Piping Code and the additional requirements of this Practice. Unless otherwise specified by the Owner's Engineer, magnetic particle examination shall be done with the wet fluorescent method using an AC Yoke.

18.4.2 Examination of fillet or branch connection weld requirements shall be examined per the methods below, following the examination percentage requirements of Table 4:

1) Fabricated branch connection welds shall be examined by either the magnetic particle or liquid penetrant method from the outside and the inside, if accessible.
2) Fillet welded joints, such as welds for slip-on flanges, socket-weld ends, etc., and seal welds shall be examined by either the magnetic particle or liquid penetrant method of the root and final weld passes, except for Category D service, which can have only the final weld pass examined.
3) Branch weld-on fittings (weld-o-lets) shall have the root and final weld passes either magnetic particle or liquid penetrant examined. Additionally, the inside of the root pass of weld-o-lets shall be visually examined to ensure full penetration.

18.5 Supplemental Inspection Requirements for Pneumatic Testing

Supplemental inspection requirements for piping which is to be pneumatically field tested are covered in EP 5-5-7.

18.6 Weld Rejection and Additional Weld Inspection

18.6.1 Completed piping, piping components, or materials containing defects originating with the Manufacturer's design, materials, or workmanship, or not in complete compliance with the requirements of the Purchase Order and referenced documents, will be rejected. Weld repair shall be per Section 20.0.

18.6.2 Discovery of such conditions after inspection and acceptance of the fabricated piping by the Purchaser does not relieve the Manufacturer of his responsibility to comply with the requirements of this Practice and the Purchase Order.

18.6.3 If radiographic or ultrasonic examination reveals a defect requiring repair, two additional welds made by the same welder immediately prior to or directly following (if first weld) the defective weld shall be inspected.

1) If both welds are acceptable, the repair shall be made and welding shall continue.

2) For each of the second group of welds revealing rejectable defects, two additional welds made by the same welder shall be examined. If a second defective weld is found, the welder shall be removed from the work force and allowed to retest after 90 days.
   - If all welds in the third group are acceptable, the second group's repairs shall be made and welding shall continue.
   - If any of the third group of welds reveal rejectable defects, 100% of the welds made by that welder shall be examined and defects repaired.

18.6.4 Additional weld inspection resulting from defective welds shall not be considered part of the original percentage required.

18.6.5 The work of each welder and each welding procedure specification shall be inspected, even if this means exceeding the required number of inspections.

18.6.6 Radiographs disclosing defects shall be followed by trace radiographs until the extent of the defect is determined, or the entire weld shall be removed and re-welded. Trace and repair radiographs shall be at the Contractor's expense.

18.6.7 All repair films and the corresponding original films of the defective weld shall both be retained for review by the Inspector.

18.6.8 (*) The Inspector reserves the right to request additional inspection in addition to that called for in this Practice. When defects are found during extra inspection, the fabricator shall bear all costs for the inspection and subsequent repair. If welds are satisfactory, the Owners will pay for the inspection according to prior agreement with the Contractor.

18.7 Hardness Tests

Brinell hardness tests of the base metal, weld metal, and the related heat affected zone (HAZ) shall be inspected per EP 10-2-3.

18.8 Positive Materials Identification

Positive materials identification (PMI) shall be per EP 15-1-4.

19.0 SHOP PRESSURE TEST

19.1 (*) Pressure hydrostatic testing of individual sections of shop fabricated piping systems is not required, unless otherwise specified by the Owner's Engineer. If specified, the pressure test shall be per EP 5-5-3. Requirements for field pressure testing of piping are stipulated in EP 5-5-3.

19.2 Each complete fired heater tube assembly (or coil configuration), including any flanges, fittings, couplings, etc., shall be hydrostatically tested. Hydrostatic test pressure shall be per ASME B31.3, but not less than 200 psig (1.4 MPaG). The hydrotest shall be per EP 5-5-3.

19.3 For jacketed pipe, hydrostatic testing of the inner pipe is required prior to assembly of the outer pipe. Requirements for hydrostatic test are stipulated in EP 5-5-3.

20.0 REJECTION AND REPAIR

20.1 Defects that are outside the limits of the codes, project specifications, or other requirements stated on the Purchase Order shall be cause for rejection. The Contractor or Manufacturer shall take such remedial action as is necessary to secure acceptance. The cost of the remedial action shall be defrayed by the contractor or vendor.

20.2 (*) Repairs of major defects in pipe, plate, or forgings require prior approval by the Owner's Engineer. Repairs of weld defects are considered major when the defect size exceeds one-half the wall thickness and the thickness of the component is over 1 inch (25 mm), or when the defect resulted in leakage during a hydrostatic or pneumatic test. The repair procedure shall be in writing and shall include information on methods used for defect removal, inspection of cavity, welding procedures, post weld heat treatment, and details of nondestructive examination of the repaired area.

20.3 (*) All welds (including weld overlays) that are found by inspection to be unsound, or that are deposited by procedures differing from those properly qualified, shall be rejected. They shall be completely removed from the equipment and replaced in accordance with approved procedure or shall be repaired, subject to approval by the Owner's Engineer.

20.4 Removal of defects by chipping, grinding, or gouging shall be done in such a manner as to avoid reducing the adjacent base material thickness. If the adjacent material thickness is reduced, it shall be restored to its original condition. Complete removal of defects shall be verified by nondestructive examination before repair is started. Repair welding shall be performed only by qualified welders using qualified procedures. On accessible joints, root defects may be repaired by welding, grinding, or both from the root side. Root defects on inaccessible joints shall be removed and re-welded. Each completed weld repair shall be verified acceptable using the same NDT method that was used to identify the defect.

20.5 When a welder's or welding operator's welding is judged unsatisfactory by the Inspector, the welder shall be removed from the work. All such welder's or operator's welding shall be inspected by nondestructive examination and welds failing the examination shall be removed or repaired as directed by the Inspector. The welder may be reassigned after additional training and the completion of satisfactory qualification tests, and only with the approval of the Inspector.

21.0 PAINTING

21.1 Following completion of inspection and testing, piping shall be painted in accordance with EP 10-3-1. Heater coils shall not be painted.

21.2 All truing rings and other temporary supports required for maintaining pipe roundness for shipping or during fabrication shall be painted yellow and tagged with the following statement:

"SHIPPING/FABRICATION DEVICE … REMOVE BEFORE UNIT START-UP."

22.0 TABLES

Table 1 – Field Joint End Preparation for Large Diameter Pipe

Type	Description
I	Field weld of two shop prepared pipe ends: See Figure 1A. This joint type can be used with or without refractory. A backing ring can be used for pipe with internal refractory lining. Truing rings are required and shall be installed within six inches (150 mm) of the end of the pipe.
II	Field weld of two shop prepared pipe ends with backing ring: See Figure 1B. This type of joint is to be used only with refractory lined pipe. This joint is designed to eliminate field installation of refractory. Truing rings are required and shall be installed six inches (150 mm) from the end of the pipe. Truing ring design and tolerance shall be per paragraph 8.3 and paragraph 8.4. This joint shall be pre-fitted and match marked in the shop.
III	Field weld of a shop prepared pipe end to an existing field joint (tie-in): See Figure 1C. Truing rings are not required; an internal spider is required for shipping.

Table 2A – Preferred Branch Connection Design

Branch Description in Terms of the Branch-to-Run Diameter Ratio	Acceptable Branch Connections and Fittings (All Temperatures and All Pressure Classes)
Run Size ≤ NPS 2 (DN 50)	Socket Weld Fitting
Run Size > NPS 2 (DN 50) and Branch Diameter > 0.5 of the Run Diameter	Welding Tee
Run Size > NPS 2 (DN 50) and Branch Diameter ≤ 0.5 of the Run Diameter	Branch Welded-on Fitting

Notes:

1) For example, Weldolets for branch sizes larger than NPS 2 (DN 50) and Sockolet or Nipolets for branch sizes less than or equal to NPS 2 (DN 50). Limitations on the use of branch welded-on fittings are specified in paragraph 11.2 of this Practice.
2) (*) With Owner's Engineer approval, additional branch connection designs shown in Table 2B may be used.

Table 2B – (*) Alternative Branch Connection Design (Requires Owner's Engineer Approval)

Branch Description in Terms of the Branch-to-Run Diameter Ratio(2)	Acceptable Branch Connections and Fittings	
	T < 800°F or Class 150,300(1)	T ≥ 800°F or Class 600, 900, 1500, 2500(1)
Run Size > NPS 2 and Branch Diameter > 0.75 of the Run Diameter	Extruded Welding Tee, Welded-in Contour Insert	Extruded Welding Tee, Welded-in Contour Insert
Run Size > NPS 2 and Branch Diameter ≤ 0.75 of the Run Diameter	Welding Tee, Extruded Welding Tee, Welded-in Contour Insert, Branch Welded-on Fitting(2)(7) Full Length Coupling(3) Fabricated Tee(4)	Welding Tee, Extruded Welding Tee, Welded-in Contour Insert, Branch Welded-on Fitting(2)(7) Full Length Coupling(3)

Notes:

1) Pressure Class is per ASME B16.5.
2) For example, Weldolets for branch sizes larger than NPS 2 and Sockolet or Nipolets for branch sizes less than NPS 2. Limitations on the use of branch welded-on fittings are specified in paragraph 11.2 of this Practice.
3) The maximum permitted coupling size is NPS 2 for run sizes greater than NPS 2.
4) Unreinforced or reinforced, depending on design requirements of ASME B31.3 or ASME B31.1, as applicable.
5) (*) Owner's Engineer approval is required for these alternative branch connection designs.
6) The connections from Table 2A were not reiterated; those connections are allowed for use within Table 2B without Owner's Engineer approval.
7) (*) For usage less than 0.75 ratio of branch/run, additional weld profile is required to improve the stress intensification factor and requires Owner's Engineer approval.

Table 3 – Pipe Joint Design Requirements

Service	Pipe Size	Weld Joint or Connection
HF Acid – Phillips Process	NPS 1-1/2 and Larger	EP 5-10-1 (Archived)
	NPS 1	
HF Acid – UOP Process	NPS 1-1/2 and Larger	EP 5-10-2
	NPS 1	
Nitric Acid and Urea	All Sizes	Butt Weld
Lube Oil and Oxygen	All Sizes	Butt Weld
Vibration/Fatigue/Cyclic Service	All Sizes	Butt Weld
Service Susceptible to Crevice Corrosion	All Sizes	Butt Weld
High Pressure Service per ASME B31.3	All Sizes	Butt Weld
Category D Service per ASME B31.3	NPS 2 and Smaller(1)	Threaded or Socket Weld
	NPS 3 and Larger	Butt Weld
All other Services	NPS 3 and Larger	Butt Weld
	NPS 2 and Smaller(2)	Socket Weld(3)

Notes:

1) Threaded joints shall be used with galvanized pipe, NPS 4 and smaller; see EP 5-2-1.
2) (*) Threaded joints may be substituted if approved by the Owner's Engineer; see EP 5-2-1.
3) (*) Threaded, seal welded joints may be substituted for specific services if approved by the Owner's Engineer; see EP 5-2-1.

Table 4 – Requirements for Non-Destructive Examination Methods

<table>
<tr><th rowspan="2">Service and Material</th><th colspan="2">Circumferential Butt Weld/Branch Weld using Radiographic Examination(1)</th><th colspan="2">Fillet/Branch Weld(4) using Liquid Penetrant/Magnetic Particle</th><th>Visual Inspection</th><th rowspan="2">Minimum Radiograph Acceptance Criteria</th></tr>
<tr><th>Shop Welds</th><th>Field Welds</th><th>Shop Welds</th><th>Field Welds</th><th>Shop and Field Welds</th></tr>
<tr><td>ASME B31.3 Piping to be Used Under Severe Cyclic Conditions</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>Severe Cyclic Conditions per ASME B31.3</td></tr>
<tr><td>Fired Heater Internal Piping (Coils), All Materials</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>Normal Fluid Service per ASME B31.3(2)</td></tr>
<tr><td>Normal Fluid Service
ASME Class 1500 and Greater, All Materials</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td rowspan="6">Normal Fluid Service per ASME B31.3(2)</td></tr>
<tr><td>All Pressure Ratings, Low Alloy (P-5B, P-5C), Titanium</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td></tr>
<tr><td>Low Alloy (P-4, P-5A)</td><td>10%</td><td>20%</td><td>100%</td><td>100%</td><td>100%</td></tr>
<tr><td>ASME Class 600 and 900, All Materials Except those which Require 100% RT</td><td>10%</td><td>20%</td><td>10%</td><td>20%</td><td>100%</td></tr>
<tr><td>ASME Class 300 and Less, Carbon Steel in AES, Stainless Steels, Nickel, Nickel Alloy, Monel, and Aluminum</td><td>10%</td><td>20%</td><td>10%</td><td>20%</td><td>100%</td></tr>
<tr><td>ASME Class 300 and Less, Carbon Steel not in AES</td><td>5%</td><td>10%</td><td>10%</td><td>10%</td><td>100%</td></tr>
<tr><td>Category D Fluid Service</td><td>5%</td><td>5%</td><td>5%</td><td>5%</td><td>100%</td><td>Normal Fluid Service per ASME B31.3</td></tr>
<tr><td>Piping Constructed to ASME B31.1
Temperature > 750°F and All Pressures</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td rowspan="3">Per ASME B31.1</td></tr>
<tr><td>Piping Constructed to ASME B31.1
Temperature ≥ 350°F and ≤ 750°F and Pressure > 1025 psig</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td><td>100%</td></tr>
<tr><td>Piping Constructed to ASME B31.1</td><td>10%(3)</td><td>20%(3)</td><td>10%</td><td>20%</td><td>100%</td></tr>
</table>

Table 4 – Requirements for Non-Destructive Examination Methods

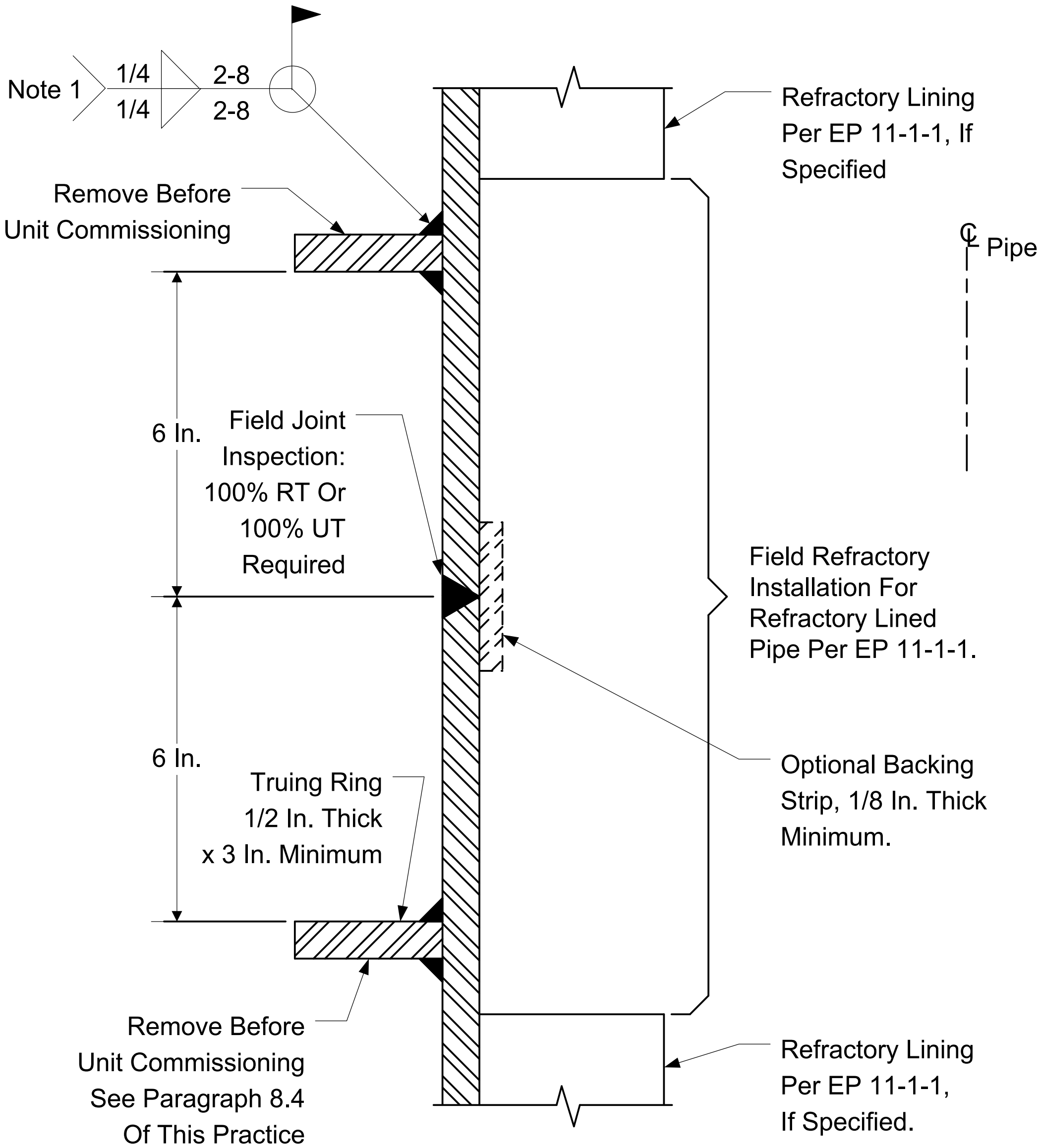
Service and Material	Circumferential Butt Weld/Branch Weld using Radiographic Examination(1)		Fillet/Branch Weld(4) using Liquid Penetrant/Magnetic Particle		Visual Inspection	Minimum Radiograph Acceptance Criteria
	Shop Welds	Field Welds	Shop Welds	Field Welds	Shop and Field Welds	
All Others						
Piping Constructed to ASME B31.4	10%(5)	10%(5)	10%(5)	10%(5)	100%	Per ASME B31.4
New-to-Existing Welds for Piping Constructed to ASME B31.3	Per EP 5-5-2, Table 3					Per ASME B31.3

Notes:

1) General notes for radiographic examination are as follows:
   a) Shop welds are welds made at the Manufacturer's place of business. Field welds are welds made at the Owner's Site.
   b) The minimum number of radiographs for Category M Fluid Service piping shall be the table values or 20%, whichever is greater. See paragraph 18.2.5 for additional information on weld inspection frequency.
2) Except that incomplete penetration is not permitted.
3) The extent of radiography in ASME B31.1 is based on pressure and temperature conditions. The percentages in this Table represent the minimum requirements.
4) Branch weld should be inspected using radiographic examination, when possible. If not, liquid penetrant or magnetic particle examination can be used. Typically, a branch weld of greater than NPS 4 can receive radiographic examination.
5) The extent of radiography in ASME B31.4 is based on location of the piping system. The percentages in this Table represent the minimum requirements.
6) Branch and fillet welds shall follow inspection methods in paragraph 18.4.2.

23.0 FIGURES

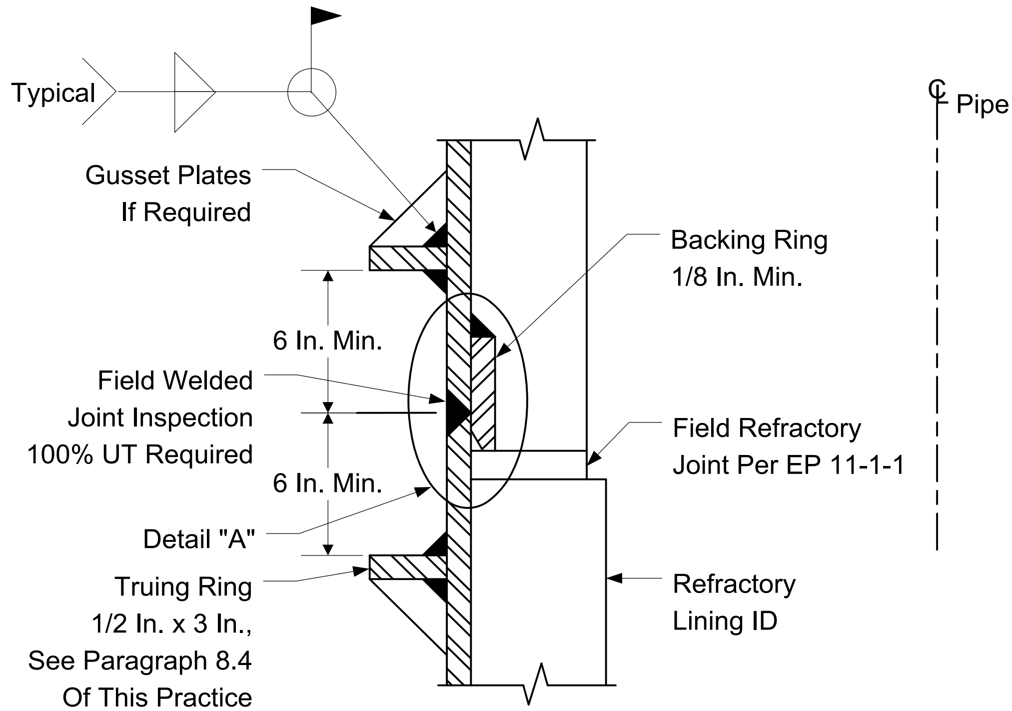
Figure 1A – Type I End Preparation

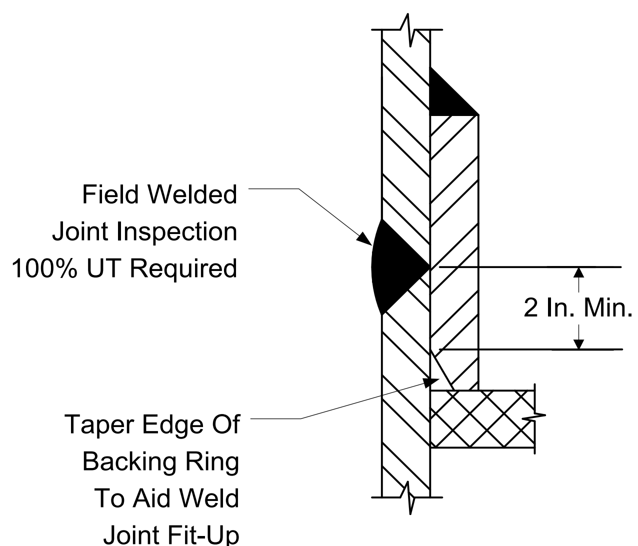


Note:

Typical weld for Type I, II, III; use continuous fillet weld if truing rings are not to be removed from refractory lined pipe. Truing rings do not have to be removed from refractory lined pipe with a design metal temperature of 400°F or less.

Figure 1B – Type II End Preparation





Detail A

Figure 1C – Type III End Preparation

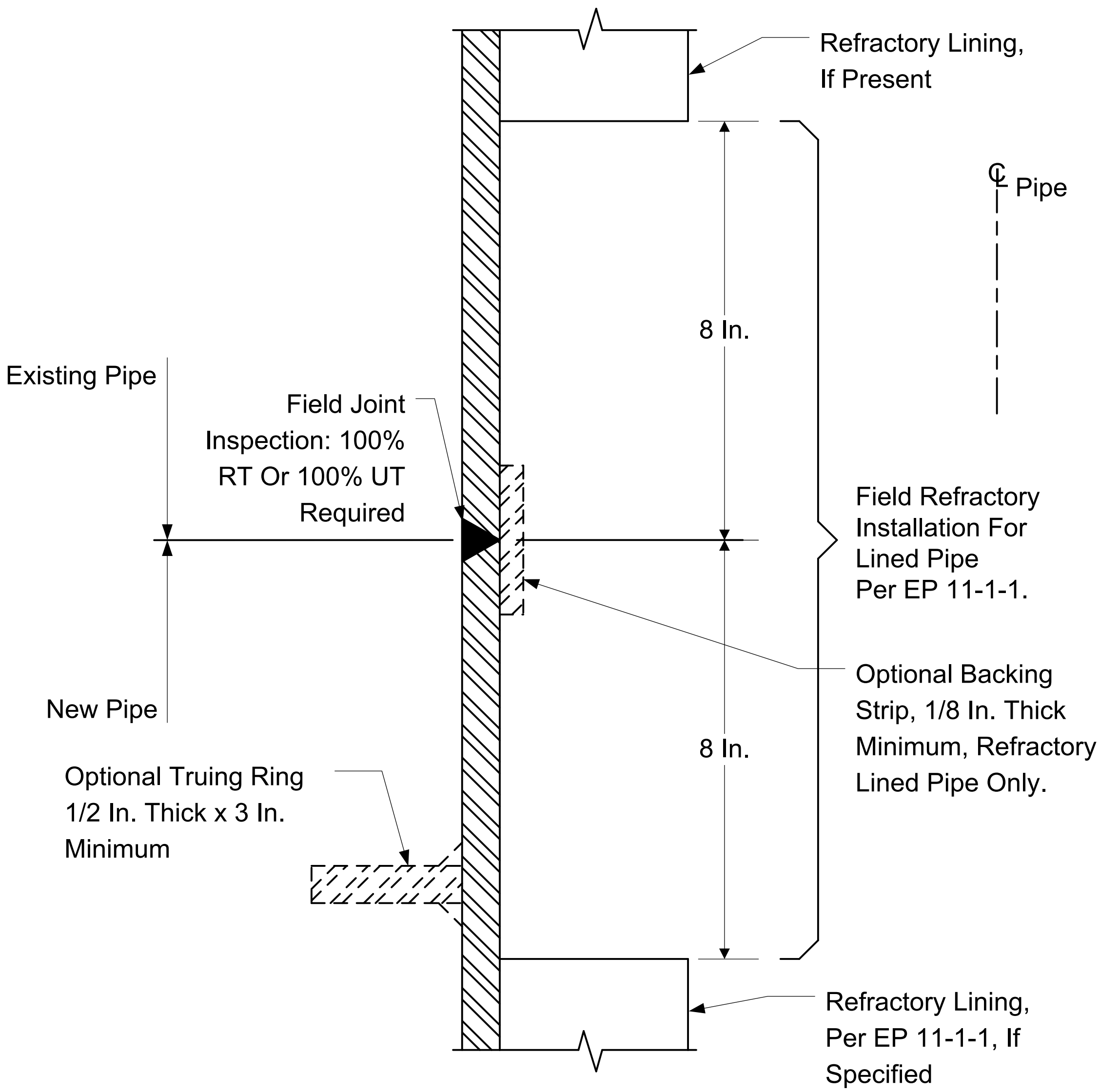


Figure 1D – Socket Weld Joint

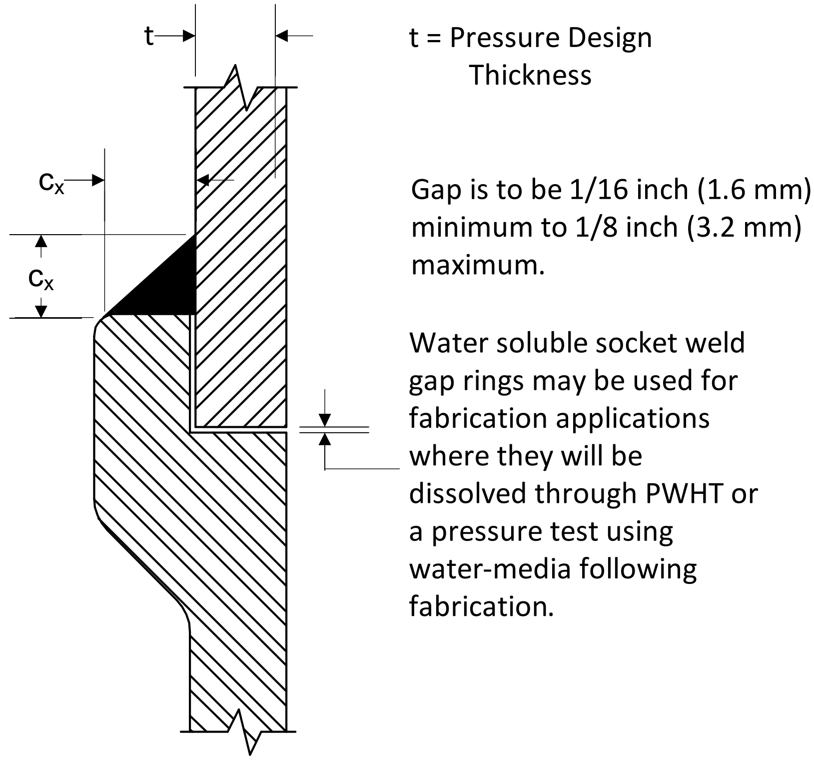
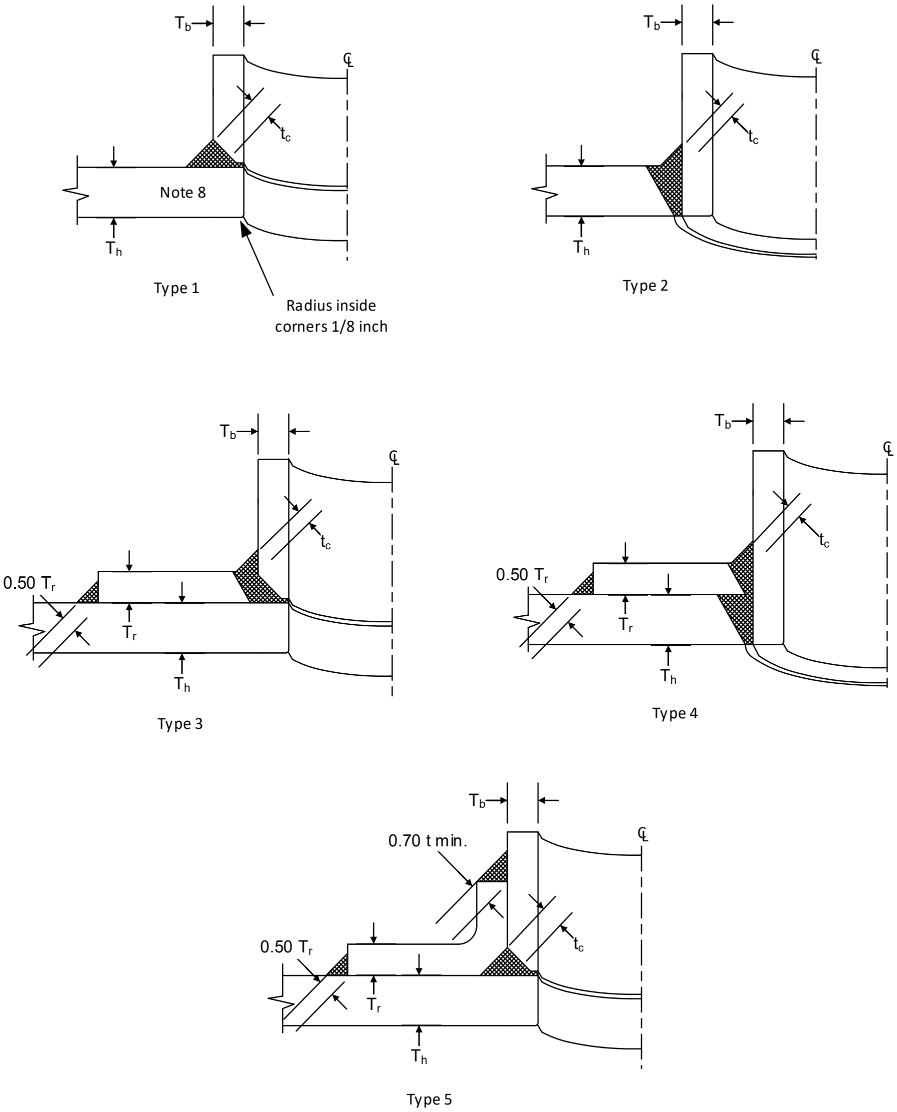


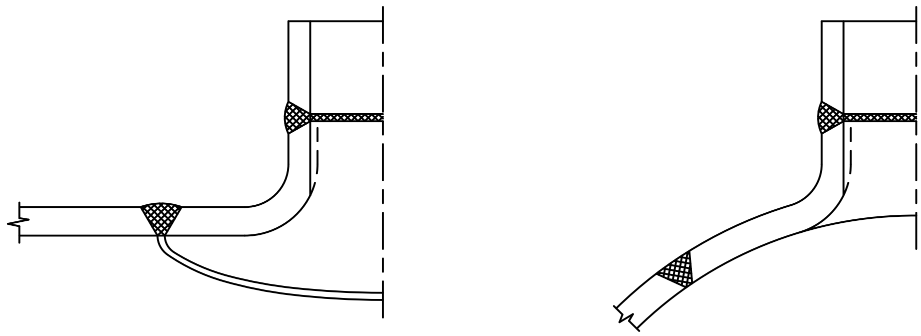
Figure 2 – Acceptable Details for Branch Attachment Welds



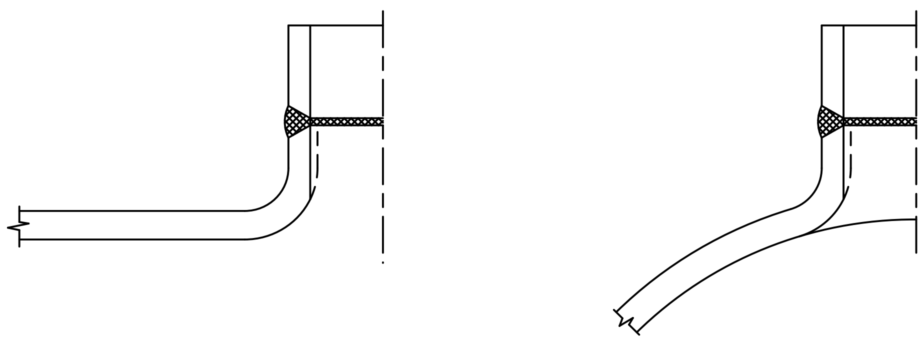
Notes:

1) (*) The use of weld attachment details Types 1, 3, and 5 requires approval of the Owner's Engineer.
2) These sketches show minimum acceptable welds. Welds may be larger than those shown here.
3) T_b – Nominal Thickness of Branch.
4) T_h – Nominal Thickness of Header.
5) T_r – Nominal Thickness of reinforcing Pad or Saddle.
6) t_c – lesser of 0.7 Tb or 0.25 inches (6 mm).
7) t_{min} – lesser of Tb or Tr.
8) For header thickness greater than 1/2 inch (13 mm), MT or PT inside surface of hole cut into header for laminations.
9) See ASME B31.3 Figure 328.5.4D.

Figure 3 – Acceptable Details for Branch Attachment Suitable for 100% Radiography



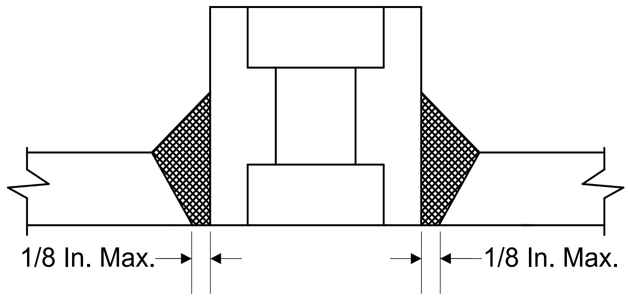
Type 1 - Contour Outlet Fitting



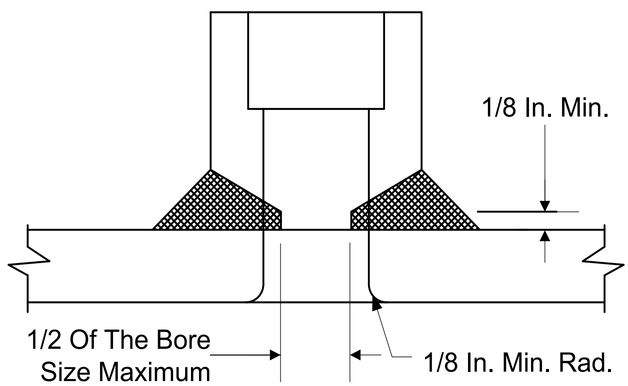
Type 2 - Extruded Header Outlet

Note:
See ASME B31.3 Figure 328.5.4E.

Figure 4 – Permissible Attachment Details for Couplings

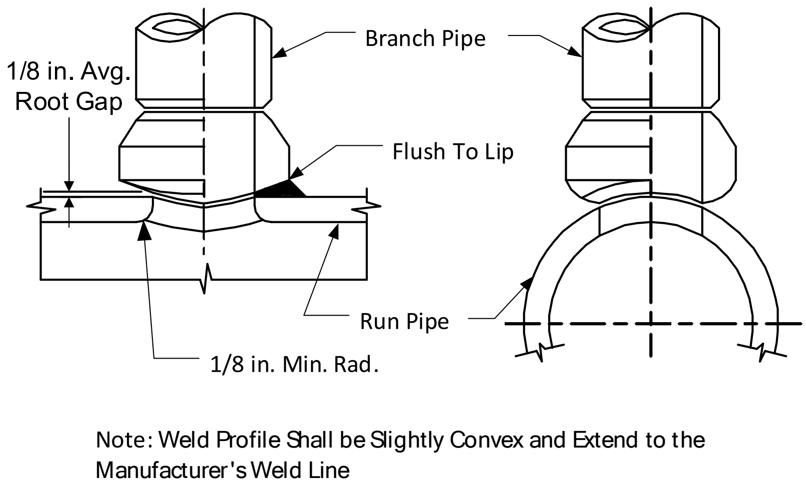


Type 1 - Set In Type Socket Weld Full Couplings
Single Or Double Bevel

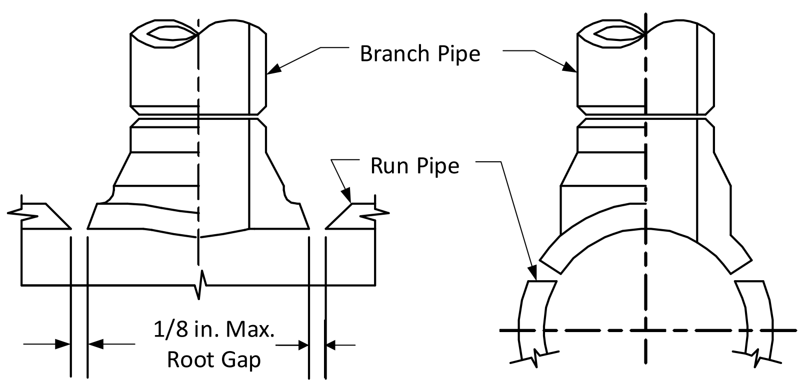


Type 2 - Set-On Drill-Thru Type Socket
Weld Couplings

Figure 5A – Acceptable Attachment Details for Integrally Reinforced Branch Connections

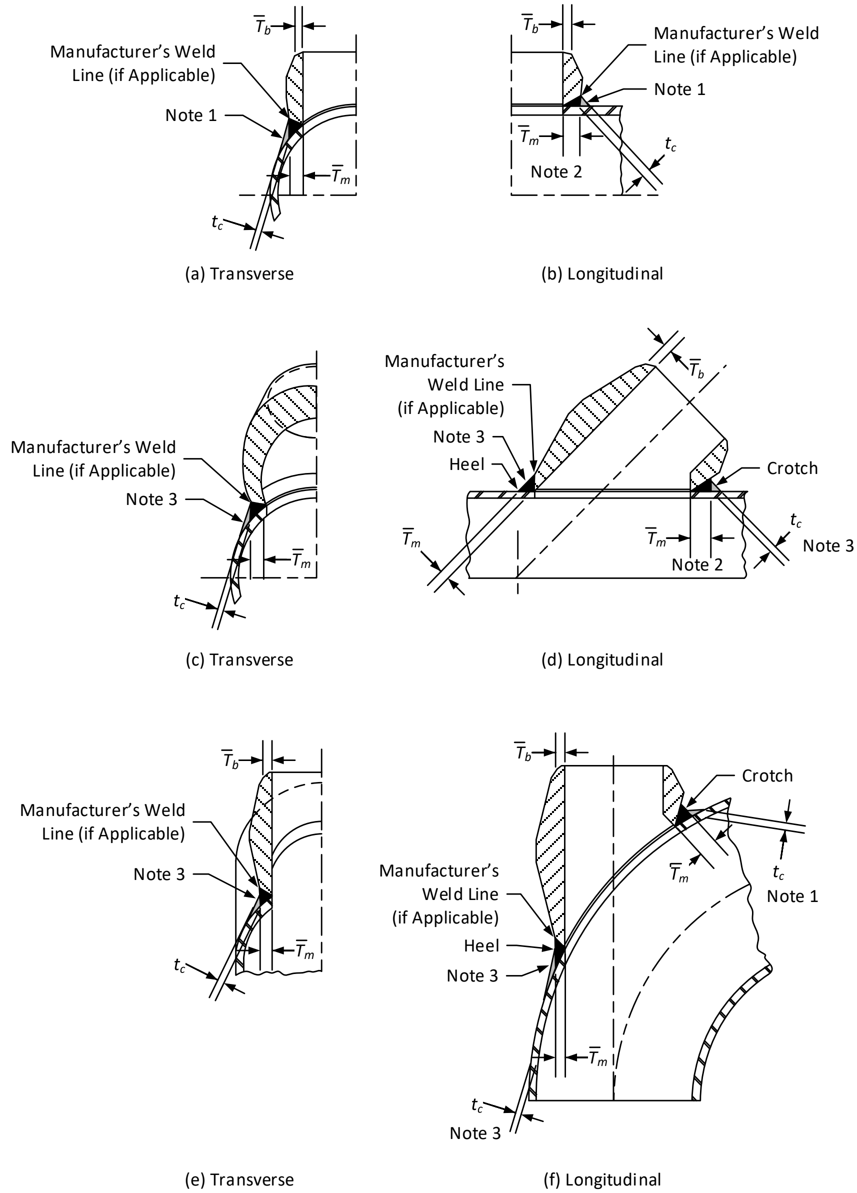


Type 1 - Set-On Type (Weldolet)



Type 2 - Set-In Type (Sweepolet)

Figure 5B – Welding Requirements for Integrally Reinforced Branch Connections



Notes:

1) Cover fillet weld shall provide a smooth transition to the run pipe with an equal leg fillet at the longitudinal section to an equal leg fillet, unequal (oblique) leg fillet, or groove butt joint at the transverse section (depending on branch connection size).
2) Heat treatment requirements shall be in accordance with paragraph 331.1.3(a) of ASME B31.3.
3) Cover fillet weld shall provide a smooth transition to the run pipe, with an equal leg fillet at the crotch in the longitudinal section to an equal leg fillet, unequal (oblique) leg fillet, or groove butt joint at the transverse section (depending on branch connection size), to nothing at the heel of the branch connection fitting in the longitudinal section.

24.0 REVISION LOG

Action	Date	Rev.	Originator	Approver	Description
Release	08/2023	99.0	The Equity Engineering Group, Inc.	The Equity Engineering Group, Inc.	Initial release.
Acceptance	12/2023	0.0	Kern Energy	Jon Keech	Client acceptance without changes.
Revision(2)	03/2024	0.1	Kern Energy	Steven Brooks	Removed SI tables and figures.
Reaffirmation	09/2024	0.2	The Equity Engineering Group, Inc.	Kern Energy(1)	This document has received a complete review and update to align with current industry standards, technologies, and best practices. Modified 7.2 and Figure 5A. Added 9.2-9.4, Section 10, 15.12, 18.1.5-18.1.6, 18.1.11-18.1.12, Table 3, and Figure 1D.
Technical	08/2025	0.3	Equity	Equity	Modified S&P Table, 1.6, 2.0, and Table 3 (Customary & SI). Removed Note 4 in Figure 5B. Minor editorial updates made throughout.
Revision	07/2026	0.4	Equity	Kern Energy(1)	Modified S&P Table, 3.1, 3.5, 3.9, 3.15., and 12.2. Added 3.6, 3.7, 3.12, and “The Why” statement to 9.7.12. Minor editorial changes made throughout.

Notes:

1) The timeframe for acceptance of this revision was exceeded per Kern Energy guidance, thus this proposed update is considered approved.
2) Company customization.